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SEMINAR AND PROJECT INITIATION COURSE

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Topic : AI-Based Deepfake Video Detection System

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Abstract

Deep learning methods such as GANs and autoencoders now make it easy to create realistic fake videos, known as deepfakes, which are misused for misinformation, identity fraud, defamation and harassment and weaken trust in video as evidence. This seminar studies an AI-Based Deepfake Video Detection System that analyses a video and decides whether it is real or fake. The system extracts frames, detects and crops faces, learns spatial features with a CNN (XceptionNet / EfficientNet), captures temporal inconsistencies with an LSTM or transformer, and classifies the video with a confidence score. Twelve research works, public datasets (FaceForensics++, Celeb-DF, DFDC) and evaluation metrics are reviewed, and the project initiation details are presented.

1. Introduction

DeepFake is a word created uisng two words “Deeplearning” and “Fake”.
DeepFake means a Fake video created using AI,
In deepfake a persons face is changed and make them say the things that they never said
It spreads Fake news,Cause fraud and damage someones reputation.
So for this reason we need a detector to detect a video if it is fake or real.

A deepfake is synthetic media (images, videos, or audio) created or altered using artificial intelligence and deep learning algorithms to make a real or fictional person appear to say or do something they never actually did

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2. Literature Survey

Sr.	Paper and Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	**FaceForensics++ — Rössler et al. (2019)** Large-scale benchmark dataset and detection study	XceptionNet and other CNN classifiers	FaceForensics++ (1,000 real videos, 4 methods)	Benchmark to train and compare detectors.	Weak on compressed videos and unseen methods.
2	**Celeb-DF — Li et al. (2020)** High-quality deepfake dataset for benchmarking	Improved synthesis; tested existing detectors	Celeb-DF v2 (590 real, 5,639 fake)	Realistic fakes to test generalisation.	Detectors score lower; celebrity subjects only.
3	**DFDC Dataset — Dolhansky et al. (2020)** Large-scale challenge dataset	Face-swap synthesis; baseline detectors	DFDC (100,000+ clips, 3,426 actors)	Large, diverse dataset for detector research.	High computing need; struggles on unseen fakes.
4	**MesoNet — Afchar et al. (2018)** Compact CNN-based detection	Meso-4 and MesoInception-4 networks	Online deepfakes and Face2Face data	Light network using mesoscopic features.	Weaker on high-quality or compressed fakes.
5	**CNN + LSTM detector — Güera and Delp (2018)** Hybrid CNN + recurrent (temporal) detection	CNN feature extractor with LSTM	Collected real and fake videos	Use frame features and temporal inconsistency.	Small training set; generalisation unsure.
6	**Face Warping Artifacts — Li and Lyu (2019)** Artifact-based detection (face warping)	CNN models (e.g., ResNet) on face regions	UADFV and DeepfakeTIMIT	Detect resolution mismatch from face warping.	Newer generators leave fewer artifacts.

Sr.	Paper and Approach Type	Technologies Used	Dataset	Key Objective	Limitations
7	**In Ictu Oculi — Li, Chang and Lyu (2018)** Physiological cue (eye blinking) detection	LRCN on eye states	Eye-blinking data by authors	Expose fakes through abnormal blinking.	Modern fakes blink naturally; eye cue only.
8	**EfficientNet + ViT — Coccomini et al. (2021)** CNN + Vision Transformer hybrid	EfficientNet with ViT; ensemble voting	DFDC and FaceForensics++	Combine local CNN and global transformer features.	High computation and data requirement.
9	**Lips Don't Lie (LipForensics) — Haliassos et al. (2021)** Mouth-movement (lip) based detection	Spatio-temporal network (lipreading)	Trained on FF++; tested on Celeb-DF, DFDC	Learn mouth-motion irregularities for generalisation.	Needs a clearly visible mouth.
10	**Multi-attentional Deepfake Detection — Zhao et al. (2021)** Attention-based fine-grained detection	Multi-attention network, texture enhancement	FaceForensics++, Celeb-DF, DFDC	Focus on local facial regions for subtle artifacts.	Complex; cross-dataset drop possible.
11	**F3-Net (Thinking in Frequency) — [Authors / year: To be confirmed]** Frequency-domain detection	Frequency-aware two-branch network	FaceForensics++	Use frequency clues for compressed fakes.	Clues weaken after resizing.
12	**Creation and Detection of Deepfakes: A Survey — Mirsky and Lee (2021)** Survey of creation and detection methods	Review of GANs, encoder-decoder models, detectors	Not applicable (survey)	Summarise creation, detection and challenges.	No new system; field changes fast.

3. Information to seminar topic

- **Technology :** Deep Learning (CNN, LSTM / Transformer) and Computer Vision
- **Main purpose :** To automatically decide whether a video is real or fake
- **Key concept :** Detectors find traces of manipulation such as blending edges, odd blinking and frame flicker
- **Approach :** Frames → MTCNN face crop → CNN features → LSTM / transformer → real / fake score
- **Main benefit :** Fast, scalable verification with a confidence score
- **Application :** Social media, news and fact-checking, law enforcement, banks (video KYC)

4. Advantages / Disadvantages

Advantages

- Fast, automatic screening of many videos
- Learns subtle artifacts that humans cannot see
- Uses both spatial (frame) and temporal (sequence) evidence
- Helps journalists, platforms and investigators trust video
- Can be retrained as new deepfake methods appear

Disadvantages

- Poor generalisation to unseen methods and datasets
- Needs large datasets and high computing power
- Compression and low quality hide artifacts
- False results can harm people or reduce trust
- Constant race between generators and detectors

5. Conclusion

Deepfakes made with GANs and autoencoders are now realistic enough to deceive viewers and threaten politics, finance, privacy and public trust. An AI-based detection system gives an automatic, scalable way to verify videos by studying each frame and how frames change over time. Modern detectors perform well on known datasets, but generalisation to new methods and robustness to compression remain open challenges. Detection is therefore a continuously evolving process that needs good datasets, regular retraining and responsible use.



Fig 2: Deepfake threat + AI detection (spatial + temporal) = Trusted video [5], [8], [12]

6. Project Initiation

The AI-Based Deepfake Video Detection project was initiated because anyone can now create convincing fake videos at low cost, and these spread quickly on social media to cause misinformation, fraud and defamation. Human viewers cannot reliably spot good deepfakes and manual forensic checking is too slow.

The project aims to build a system that extracts frames, detects faces with MTCNN, learns features with a CNN, analyses temporal behaviour with an LSTM / transformer and classifies the video as real or fake with a confidence score.

6.1 Description of Problem Statement

Deepfake videos are easy and cheap to create and spread widely on social media before they are noticed.

Human viewers and manual forensic checks cannot keep up with high-quality fakes.

Existing detectors work well only on trained datasets and fail on compressed or new fakes. A system using spatial and temporal evidence is needed.

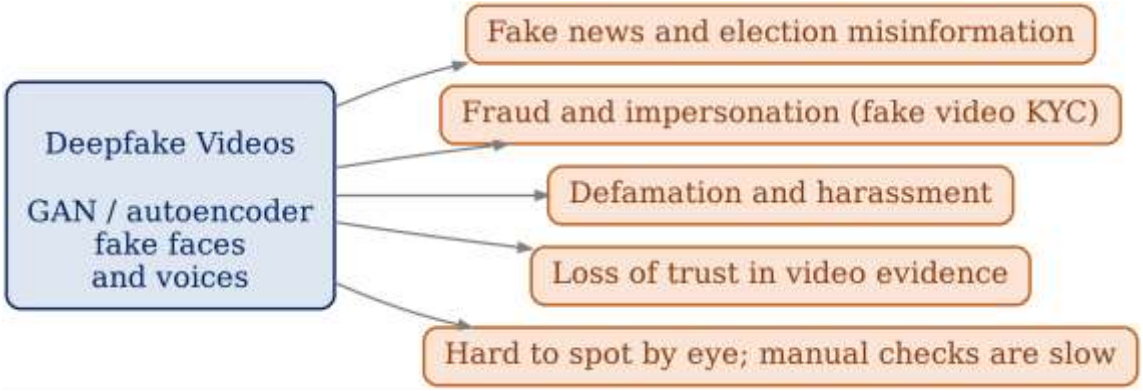


Fig 3: Problems caused by deepfake videos [3], [12]

6.2 Scope and Objectives

Objectives

- Study how deepfakes are created and the artifacts they leave.
- Review existing research on deepfake video detection.
- Build a pipeline for frame extraction, face detection and pre-processing.
- Classify a video as real or fake using CNN and LSTM / transformer.
- Evaluate with accuracy, precision, recall, F1-score and AUC .
- Show the label and confidence score in a simple form to the user.

Scope

- Detects face-based deepfake videos; prototype on public datasets.
- Uses Python, PyTorch / TensorFlow, OpenCV and MTCNN.
- Audio-only deepfakes and live-broadcast detection are out of scope.
- Helps users, platforms and investigators verify videos.

6.3 Stakeholders

- **End Users / General Public:** Upload videos to check real or fake and avoid misinformation [12].
- **Social Media Platforms / Media Houses:** Flag fake videos and verify content before sharing [3].
- **Law Enforcement / Legal Authorities:** Use reports as evidence in fraud and defamation cases [12].
- **Government / Regulatory Bodies:** Policy making, election integrity and national security [12].
- **Banks / Corporates / Victims:** Prevent fraud such as fake video KYC and protect identity [3].
- **Developers and Project Guide:** Build, test and evaluate the system.

6.4 Seminar Execution Plan

1. **Current Scenario:** Deepfake videos spread quickly on social media and messaging apps

2. **Deepfake Challenges:** Misinformation, fraud, defamation; humans cannot spot good fakes

3. **Existing Approaches:** CNN, LSTM, ViT, lip, attention and frequency based detectors

4. **Research Gap:** Detectors work on trained datasets but fail on compressed or unseen fakes

5. **Proposed System:** Frame extraction + MTCNN + CNN + LSTM / transformer with confidence score

6. **Future Impact:** Reliable video verification and more trust in digital media

7. **Future Work:** Real-time detection, audio deepfakes and better generalisation

6.5 Platform / Equipment / Resources

Hardware Requirements:

- Computer / laptop (16 GB RAM recommended)
- NVIDIA GPU (minimum 6 GB VRAM) for training
- SSD / hard disk (minimum 512 GB) for datasets
- Webcam / smartphone (optional) and Wi-Fi

Software Requirements:

- Operating system: Windows 11 / Ubuntu
- Python 3.x with PyTorch / TensorFlow-Keras
- OpenCV (frames) and MTCNN (face detection)
- NumPy, Pandas, scikit-learn, Matplotlib
- VS Code / Jupyter Notebook / Google Colab (free GPU)

Other Resources:

- Datasets: FaceForensics++, Celeb-DF, DFDC
- Streamlit / Flask demo interface (optional); internet connection

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Thank you